

Main Drivers of Rental Prices in Kuala Lumpur and Selangor

ADS1001 Group 3:

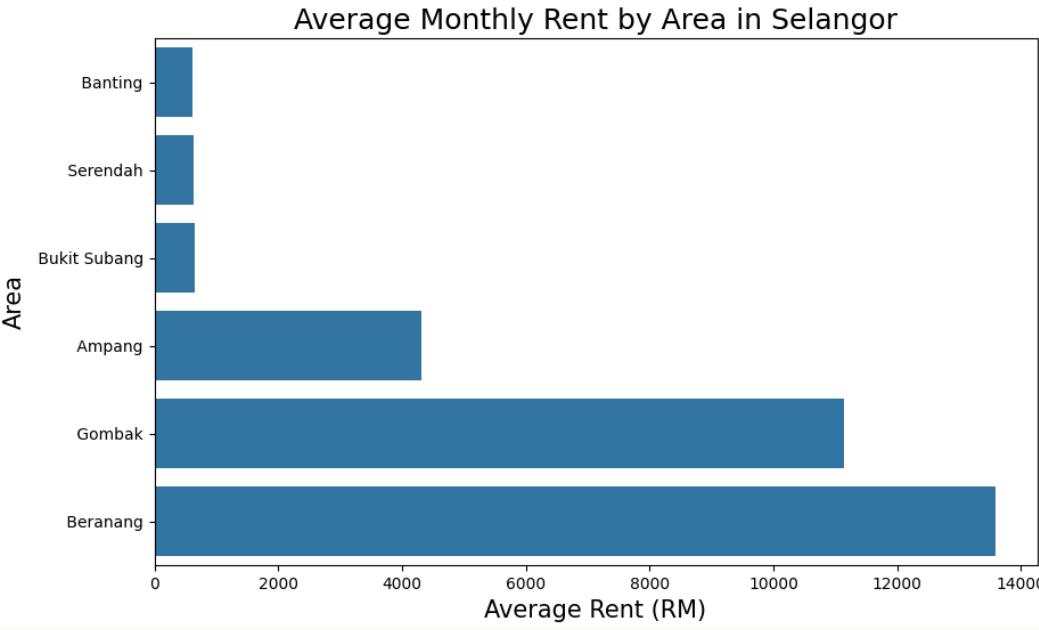
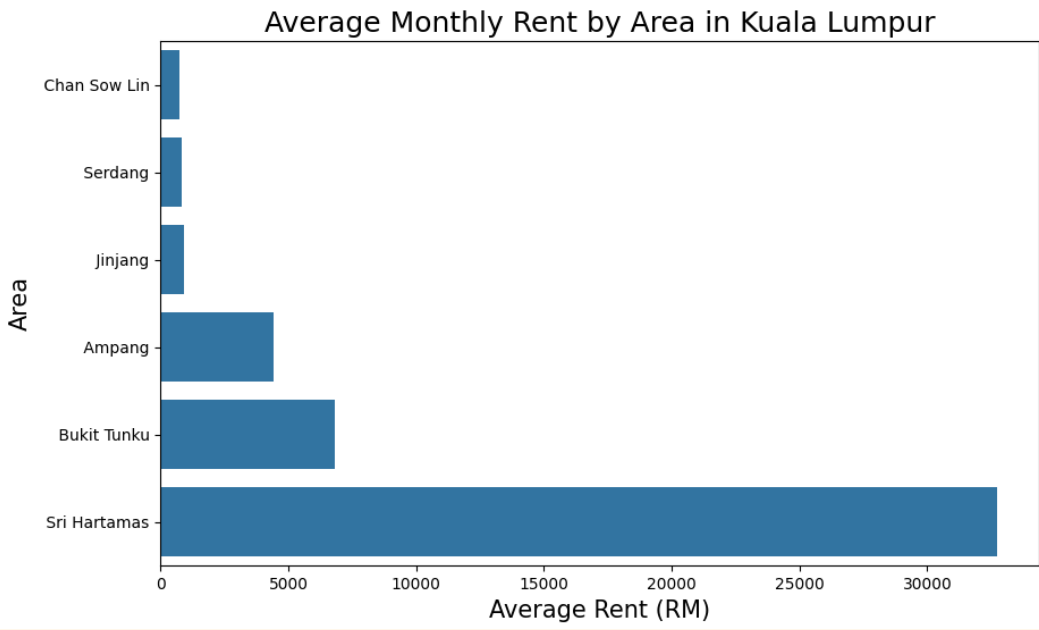
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This project uses real-world housing data in Kuala Lumpur (KL) and Selangor to explore key factors that influence rental prices. By examining these trends, our main goal is to understand what makes a property’s rent fair and valuable to tenants, landlords, and investors.



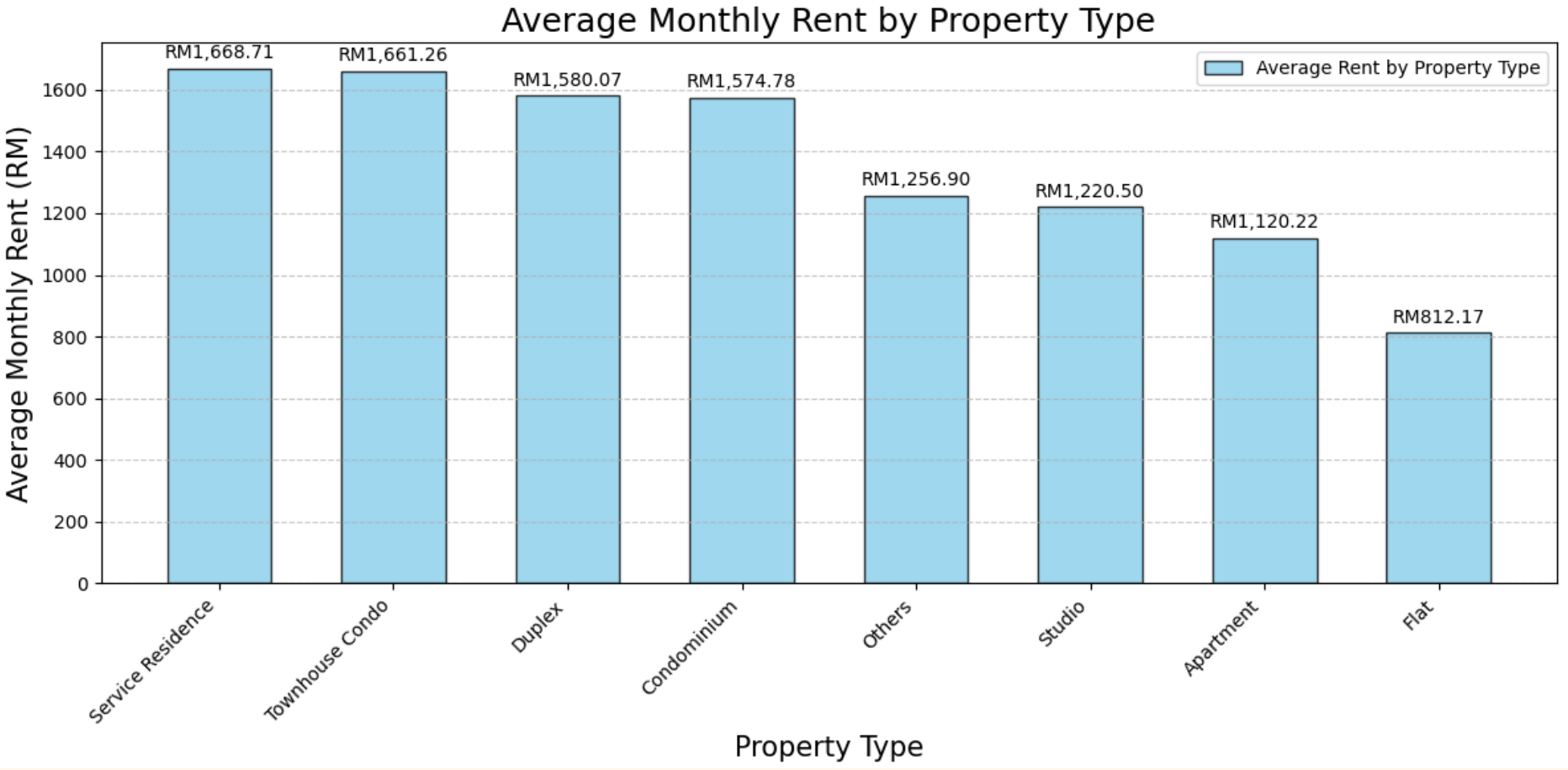
1. Area and Cities



Average Monthly Rental in KL:
Highest: Sri Hartamas (RM 3,2751.90)
Lowest: Chan Sow Lin (RM 720.00)

Average Monthly Rental in Selangor:
Highest: Beranang (RM 1,3580.00)
Lowest: Banting (RM 611.67)

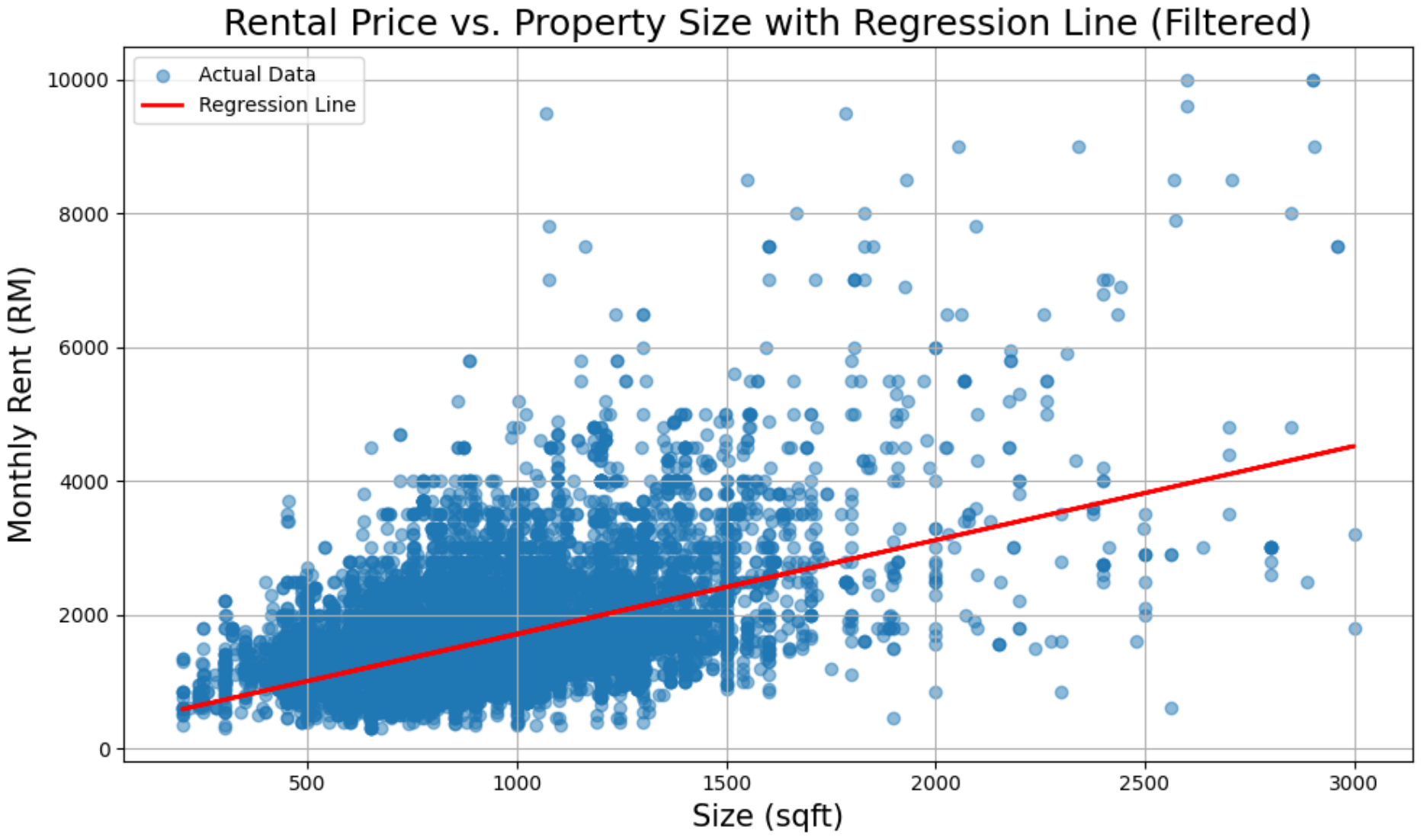
2. Property Type



Highest Average Monthly Rents:
1.Service Residences (RM 1,668.71)
2.Townhouse Condo (RM 1661.26)
3.Duplex (RM 1,580.07)

Lowest Average Monthly Rents:
1.Flat (RM 812.17)
2.Apartment (RM 1,120.22)
3.Studio (RM 1,220.50)

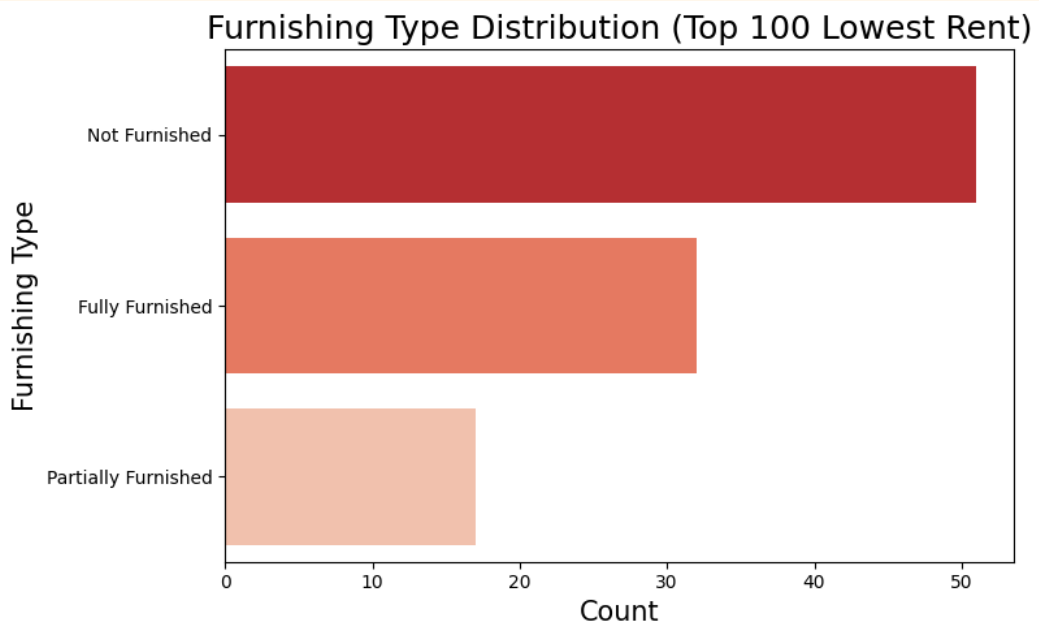
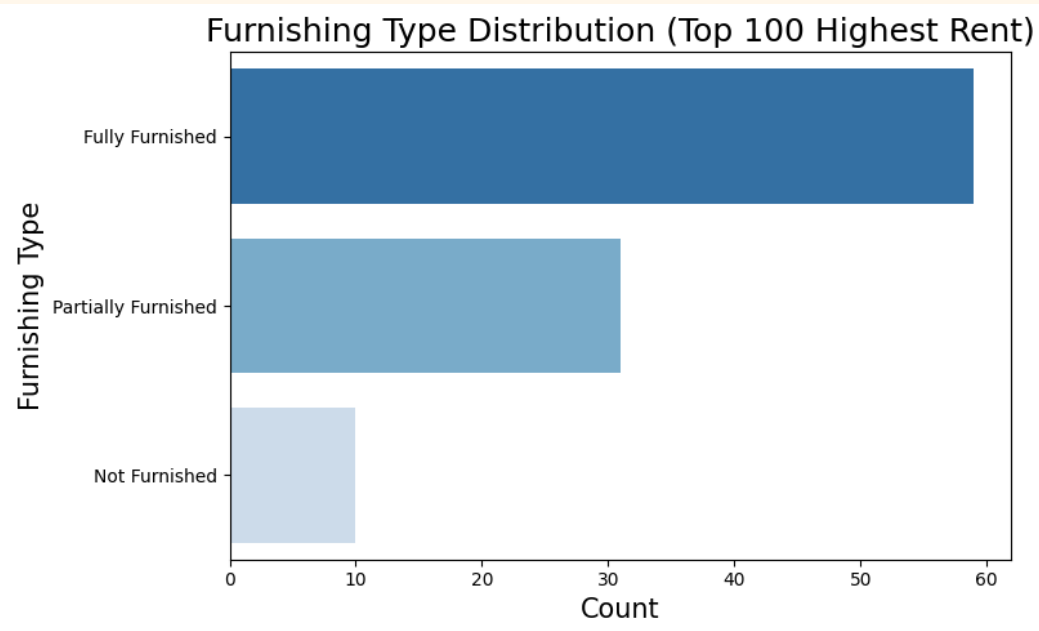
3. Property Size



- Moderate positive relationship between size and rent (0.5032 correlation coefficient)
- Regression slope: 1.4049 (Rent increases by approx. RM 1.40 per additional square foot)
- R-squared: 0.2532 (25% of rent variation)

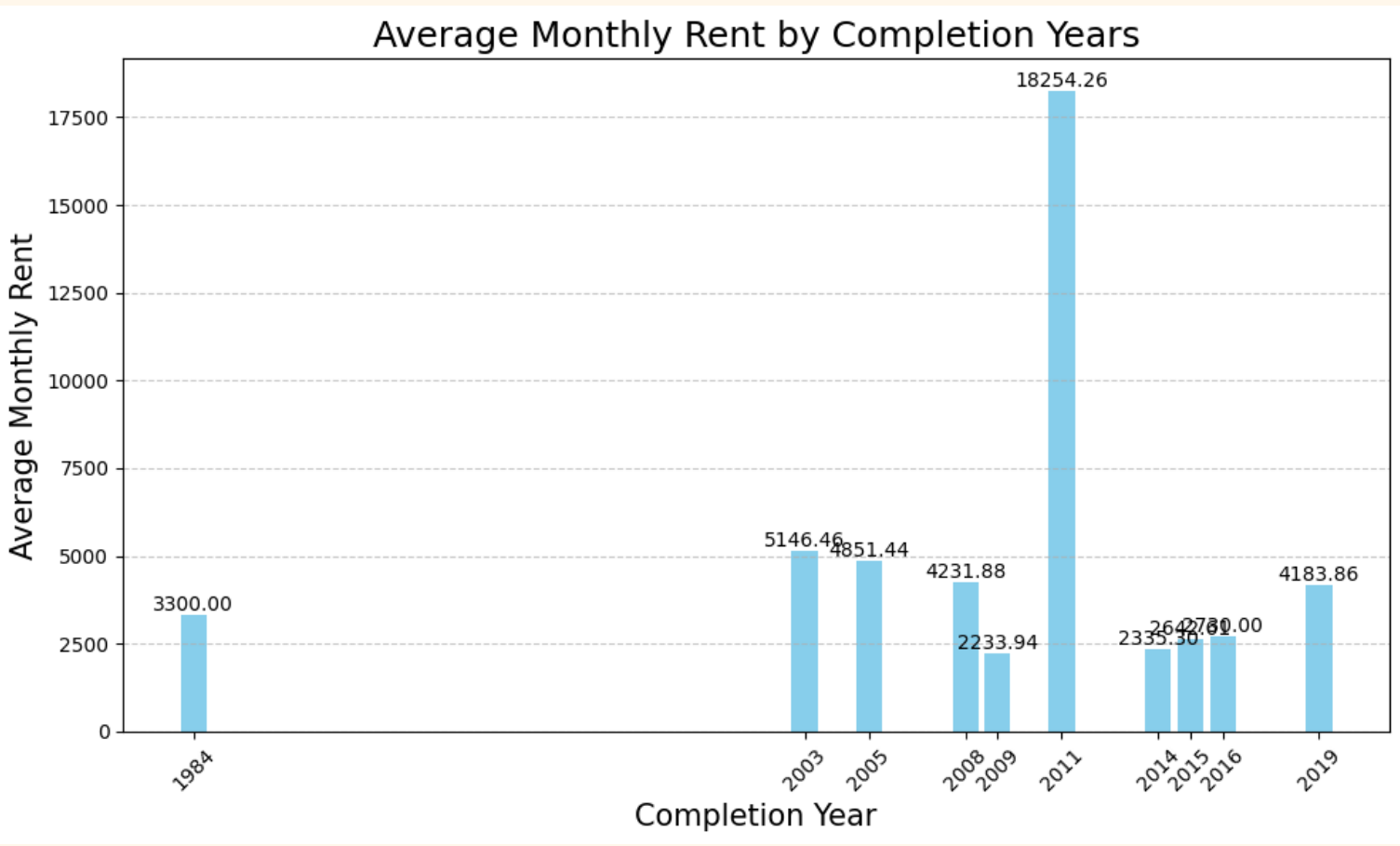


4. Furnishing Status



- Fully furnished units have higher rents
- Majority of the top 100 most expensive rentals are fully furnished
- The top 100 cheapest rentals are mostly partially or not furnished

5. Age of Properties



- Highest Average Monthly Rental: Year 2011 (RM 18,254.26)
- Lowest Average Monthly Rental: Year 2009 (RM 2,233.84)
- Properties built before 2015 generally have higher monthly rents

6. Facilities Offered

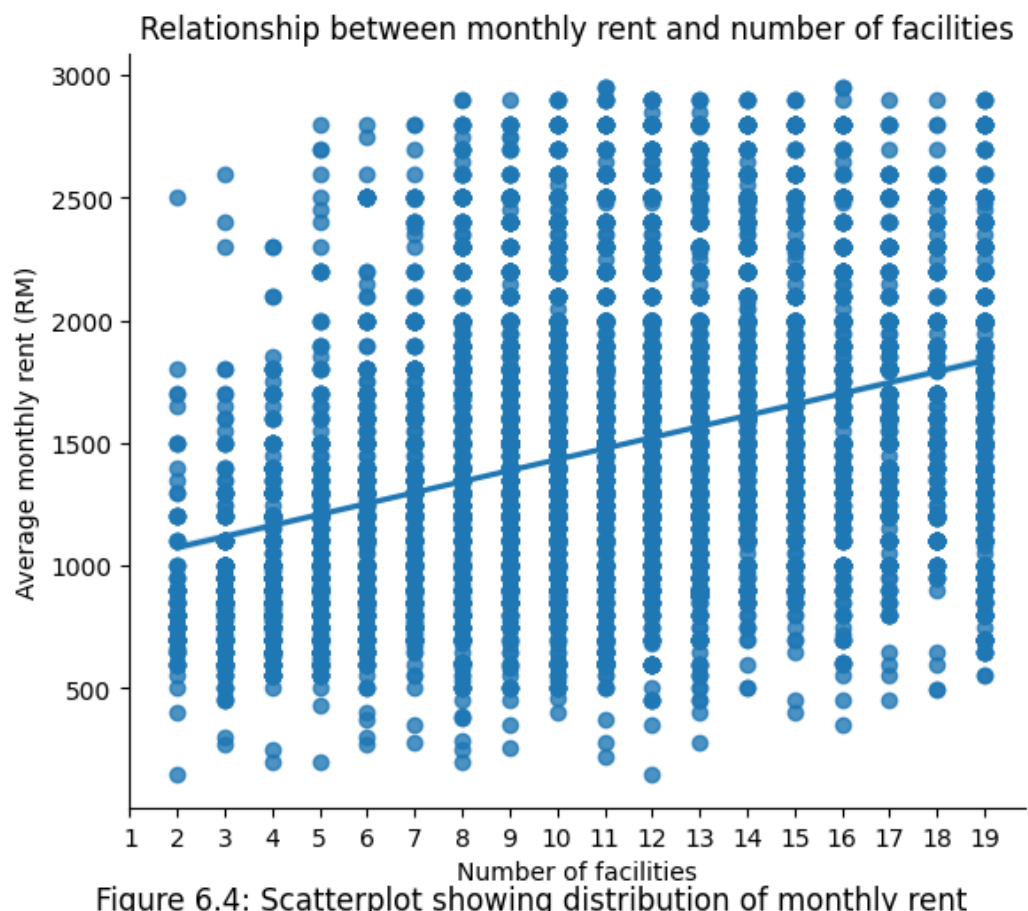


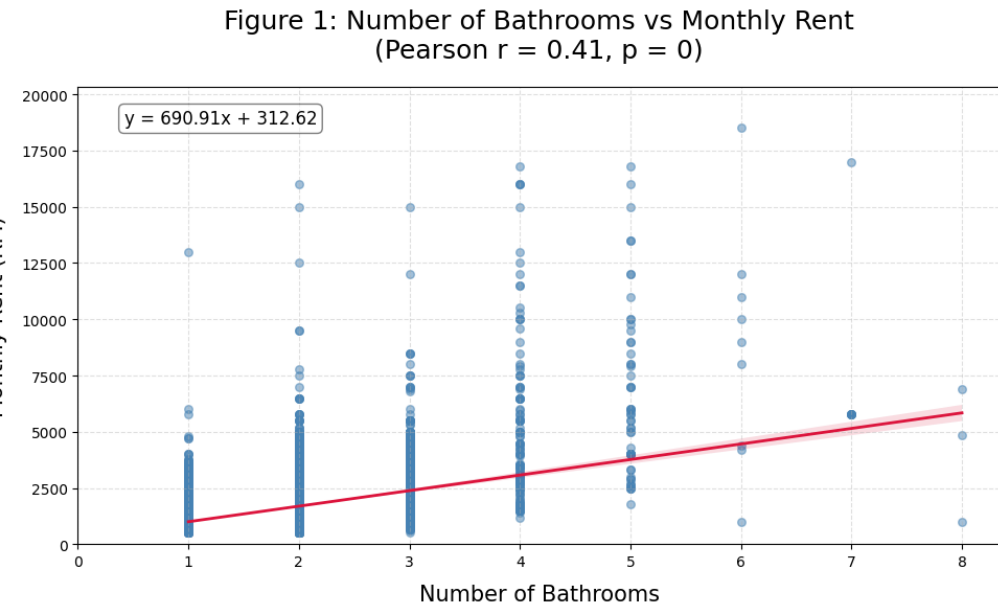
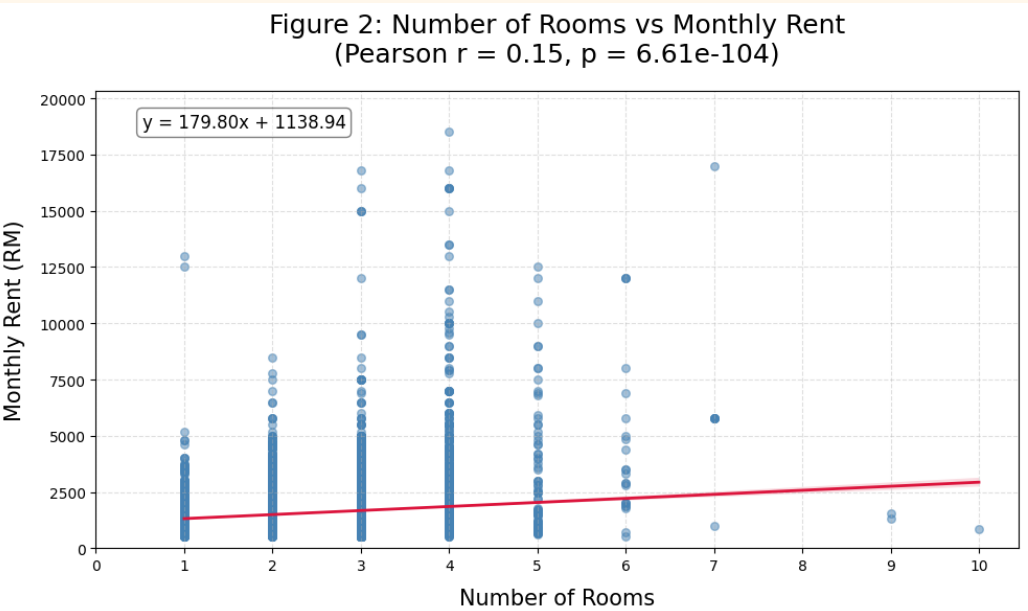
Figure 6.4: Scatterplot showing distribution of monthly rent categorized by the number of facility counts (including any additional facilities).



Figure 6.6: Boxplot showing distribution of monthly rent categorized by the number of parking provided.

- More facilities = Higher Rent**
 - Move convenient, higher demand, higher management fees
- More parking spaces = Higher Rent**
 - Larger unit, Limited parking spaces at busy areas, security fees

7. Number of Rooms and Bathrooms



- Number of Rooms VS Monthly Rent**
 - Weak but significant positive correlation
 - Rent increases by ~ RM 179.80 / additional room
- Number of Bathrooms VS Monthly Rent**
 - Moderate positive correlation
 - Rent increases by ~ RM 690.91 / additional bathroom

Polynomial Regression Fit for Size VS Rent at Degree 3

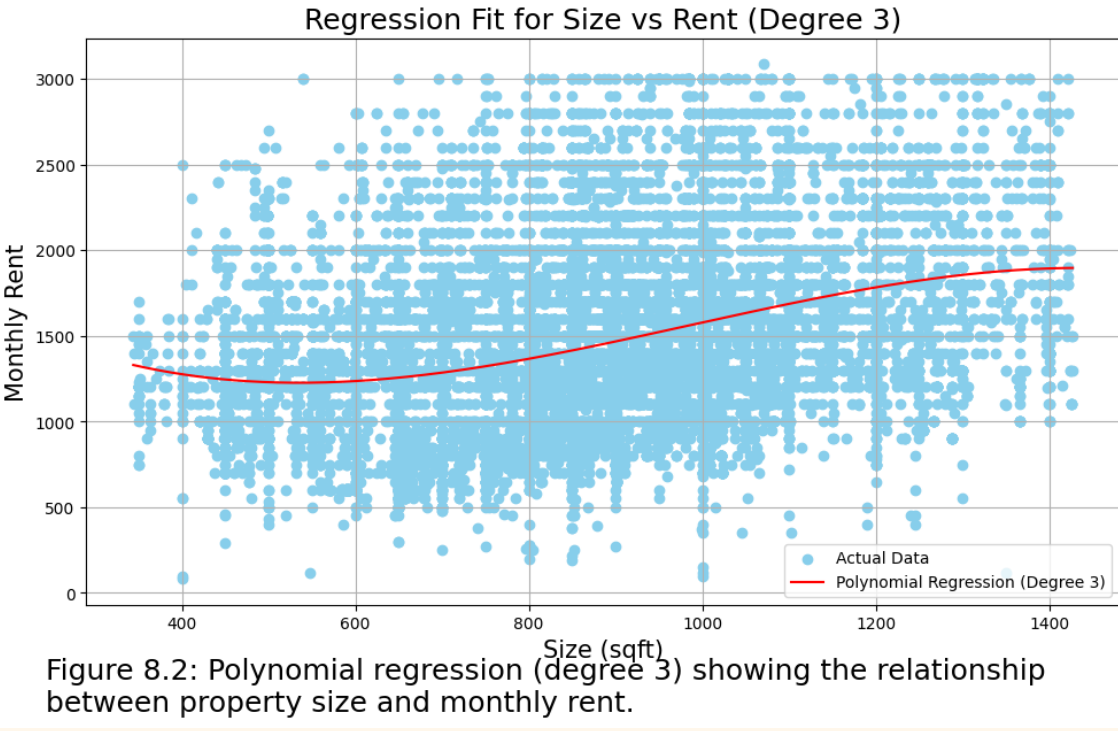


Figure 8.2: Polynomial regression (degree 3) showing the relationship between property size and monthly rent.

Blue dots = Actual data
Red line = Predicted monthly rent

Train $R^2 = 0.564$, Test $R^2 = 0.545$
~56% of rent variation
~54% for new (unseen) data

Bigger properties cost more, but rent doesn’t rise evenly with size. Other factors affect the rental price as well, making the relationship less smooth.

Features:
Property type, number of rooms, size, furnished status, number of facilities, city



From this study, shown that rental prices are affected by a diverse range of factors. By understanding the 'why' behind rental costs, this study helps tenants, landlords or housing agents navigate the housing market with greater clarity and confidence.